Activity - 04

Part A

	Violin/Group	Skew	mean vs median
Chart-1	Brand Y	left	mean < median
Char-1 Fuel efficiency	Brand V	right	mean > Median
Chart-1 Fuel efficiency	Brand X	symmetric	mean = median
Chart-2 Waiting time	Ward C	Left	mean < median
Chart-2	Ward D	symmetric	Mean = median
Chart-2	Ward B	Right	mean > median
Chart-3 Resting heart	Club E	Left	mean < median
Chart-3	Club D	right	mean > median
Chart-3	Club A	Symmetric	mean = median
Chart-4 sleep tonight	Year 2	symetric	mean = median
	Year 1	Left	mean < median
	Staff	right	Mean > median

Part-B

B1 Chart 4 - Year 1

The tail points to the left . so year 1 is left-skewed. The median is the better measure of typical sleep because it is less effected by the long tail and extreme values

B2 One-line ggplot2

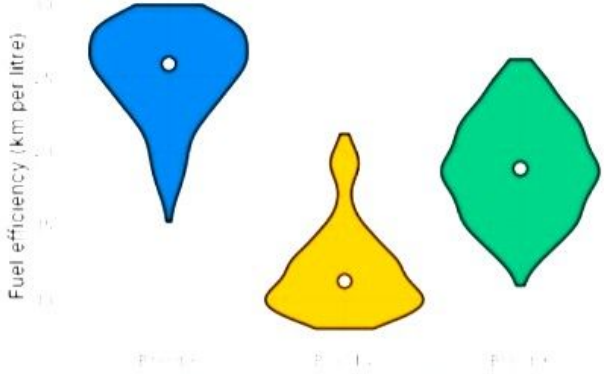
```
ggplot(df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point")
```

DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot C

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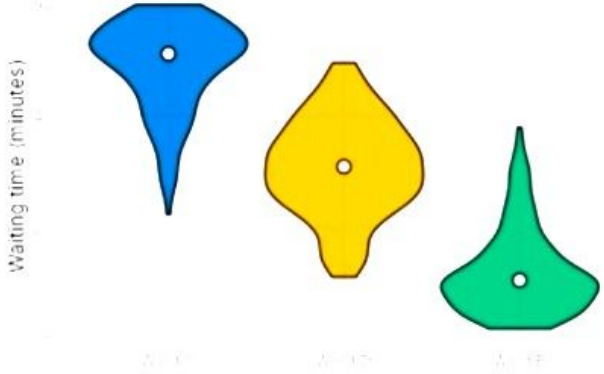
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Fuel efficiency (km per litre)



Violin	Skew (circle one)	Mean vs Median (circle)
Brand Y	Right / Left / Symm.	Mean > / < / ~ Median
Brand V	Right / Left / Symm.	Mean > / < / ~ Median
Brand X	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Waiting time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle)
Ward C	Right / Left / Symm.	Mean > / < / ~ Median
Ward D	Right / Left / Symm.	Mean > / < / ~ Median
Ward B	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Resting heart rate (beats per min)

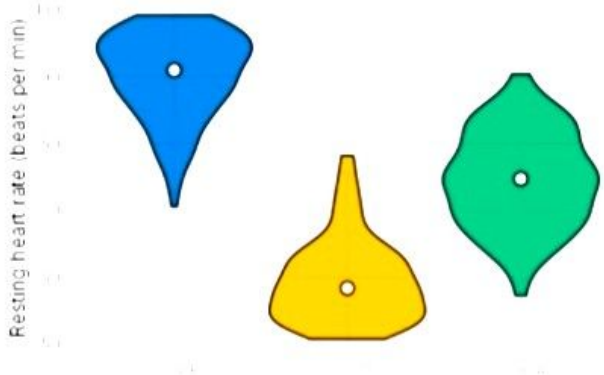
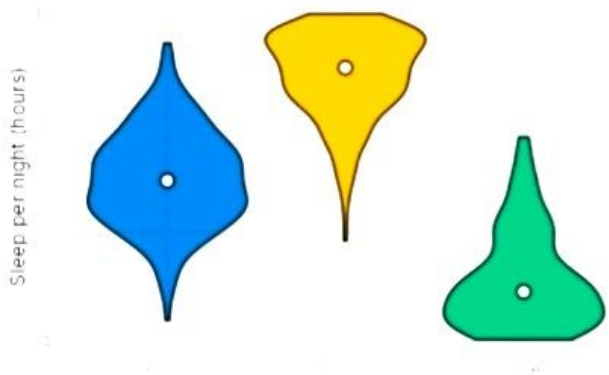


Chart 4: Sleep per night (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Club E	Right / Left / Symm.	Mean > / < / ~ Median
Club D	Right / Left / Symm.	Mean > / < / ~ Median
Club A	Right / Left / Symm.	Mean > / < / ~ Median

Violin	Skew (circle one)	Mean vs Median (circle)
Year 2	Right / Left / Symm.	Mean > / < / ~ Median
Year 1	Right / Left / Symm.	Mean > / < / ~ Median
Staff	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 4, violin "Year 1" is not symmetric. Towards which side is its tail, and which is the better measure of a typical sleep per night here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]